

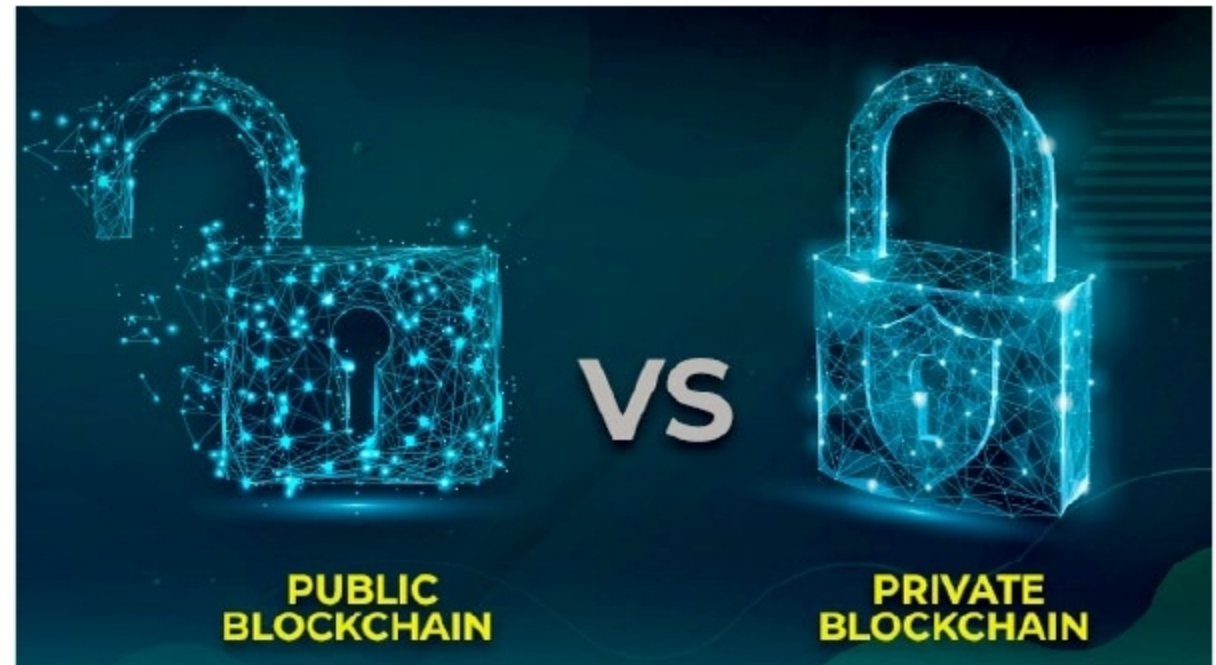
Public Blockchains vs Private Blockchains: How and When to Use Them

The blockchain platform is most popular in transaction handling and workflow related use cases, across industries. This is due to it being decentralised and the immutable nature of its transaction records stored in the ledger. Based on usability and access level architecture, blockchain platforms are classified as either public or private. This article explains the basics of public and private blockchains. It also compares the permissioned and permissionless ledgers in these platforms.

The blockchain platform is popular due to its decentralised nature and how secure the transactions on it are. There are many blockchain platforms like Ethereum and Hyperledger, to name a few, that are popular. Since most of these platforms use cryptocurrency, transaction processing is very fast compared to that in traditional banks due to the nature of its connected network.

With this background, one has to decide which blockchain platform is suitable for a particular transaction, when defining a blockchain based solution. A blockchain platform can be public or private.

Public blockchain platforms like Bitcoin or Ethereum have a network



architecture whereby anyone can participate and access the data in the network. They are permissionless blockchain platforms that anyone can read, write or even participate in. The data in a public blockchain platform is highly secure. It cannot be tampered with or changed once it is validated in the blockchain and is, at the same time, decentralised in nature so no one person can control the transaction easily.

Public blockchain architecture

In a public blockchain, participants can be even anonymous to the external world (but remain as registered users internal to the network) and they can access, change or validate the data in the transaction processing of the network, as shown in Figure 1.

When you are looking for a blockchain platform, which can handle open, immutable transactions and a distributed ledger, you can choose a public blockchain like Ethereum or Bitcoin. Even when the public

blockchain is an open network, it remains decentralised, and assures transparent and secure transaction processing in the network.

Private blockchain

A private blockchain is a permissioned blockchain platform in which a workflow is followed, on who can participate and approve the transactions. It is more highly secured than a public blockchain platform due to its secure nature. HyperLedger, Ripple or Corda are some examples of private blockchain platforms.

In a private blockchain network, a group of owners handles user permissions for the network like viewers, participants, approvers and administrators. A workflow of validation and approval is followed to ensure no unethical transaction happens in the network. When you are looking for a blockchain platform for faster transactions and a scalable architecture, you can choose a private one.

